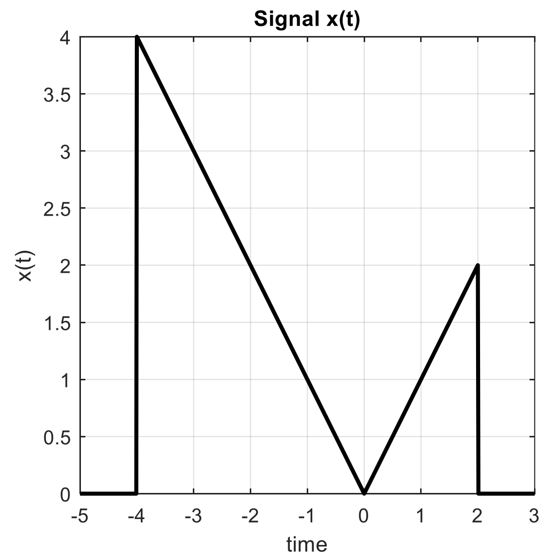
EE2008: FINAL EXAM SOLUTION MANUAL (SPRING 2025)

CLO # 1: Express different types of signals and systems.

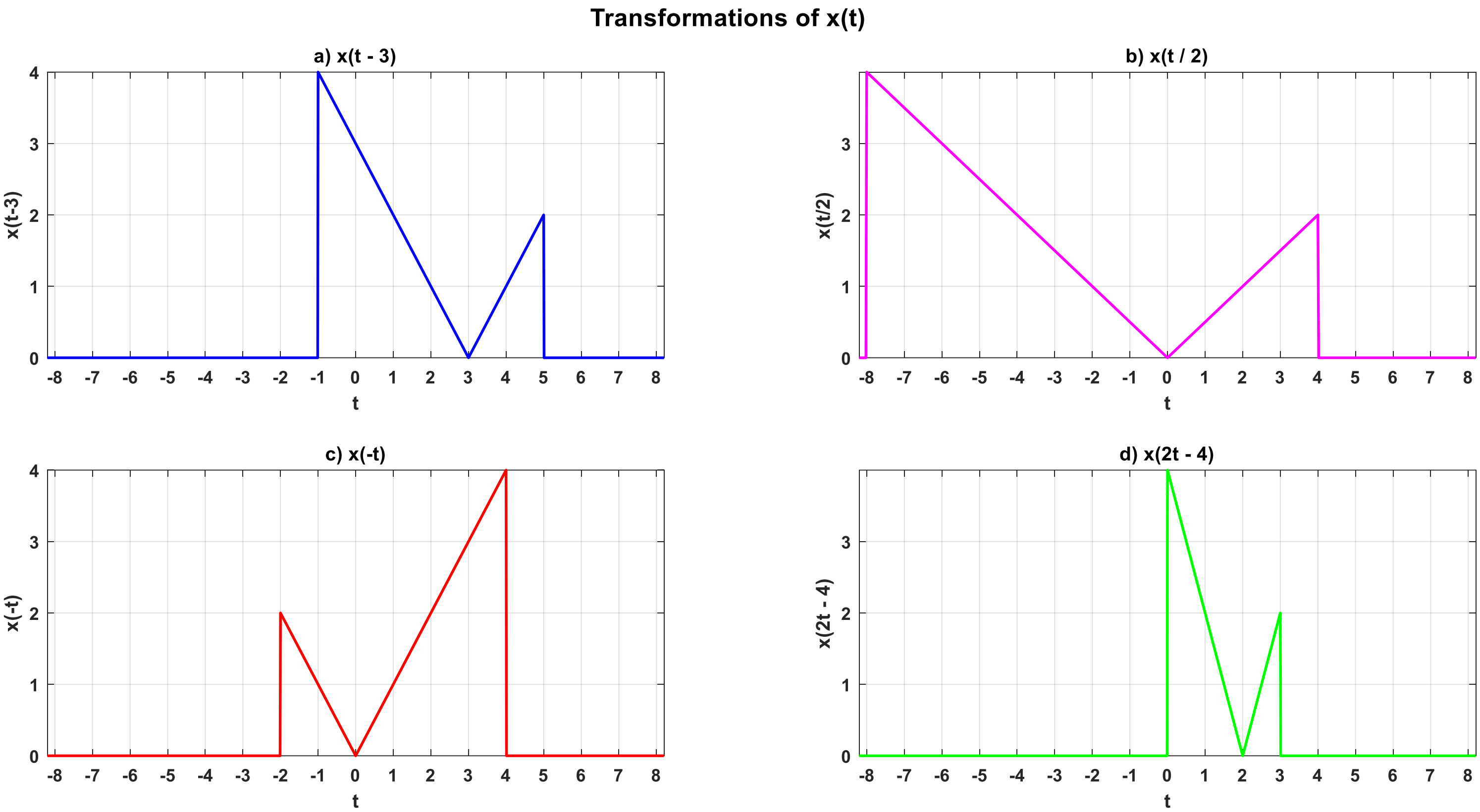
Q1: A continuous-time signal x(t) is shown in Fig-1 below. **Compute** and label carefully each of the following signals: **[20]**

a) $x(t-3)$
b) $x(t/2)$
c) $x(-t)$
d) $x(2t-4)$



Solution:

Signals	Operations
x(t-3)	Time shifting : towards right by 3
x(t/2)	Time Expansion
x(−t)	Time Reversal
x(2t-4))	Time Compression & Time shifting



CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Consider the following two signals x(t) and h(t) as shown in Figure 2. **Apply** graphical convolutional technique to compute the convolution of these two signals. **[20 marks]**

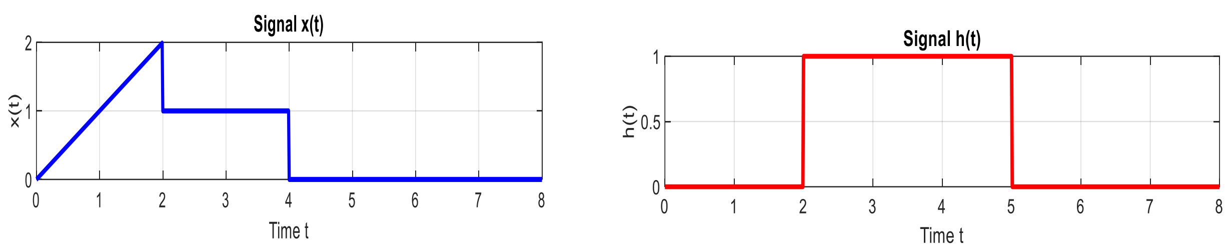
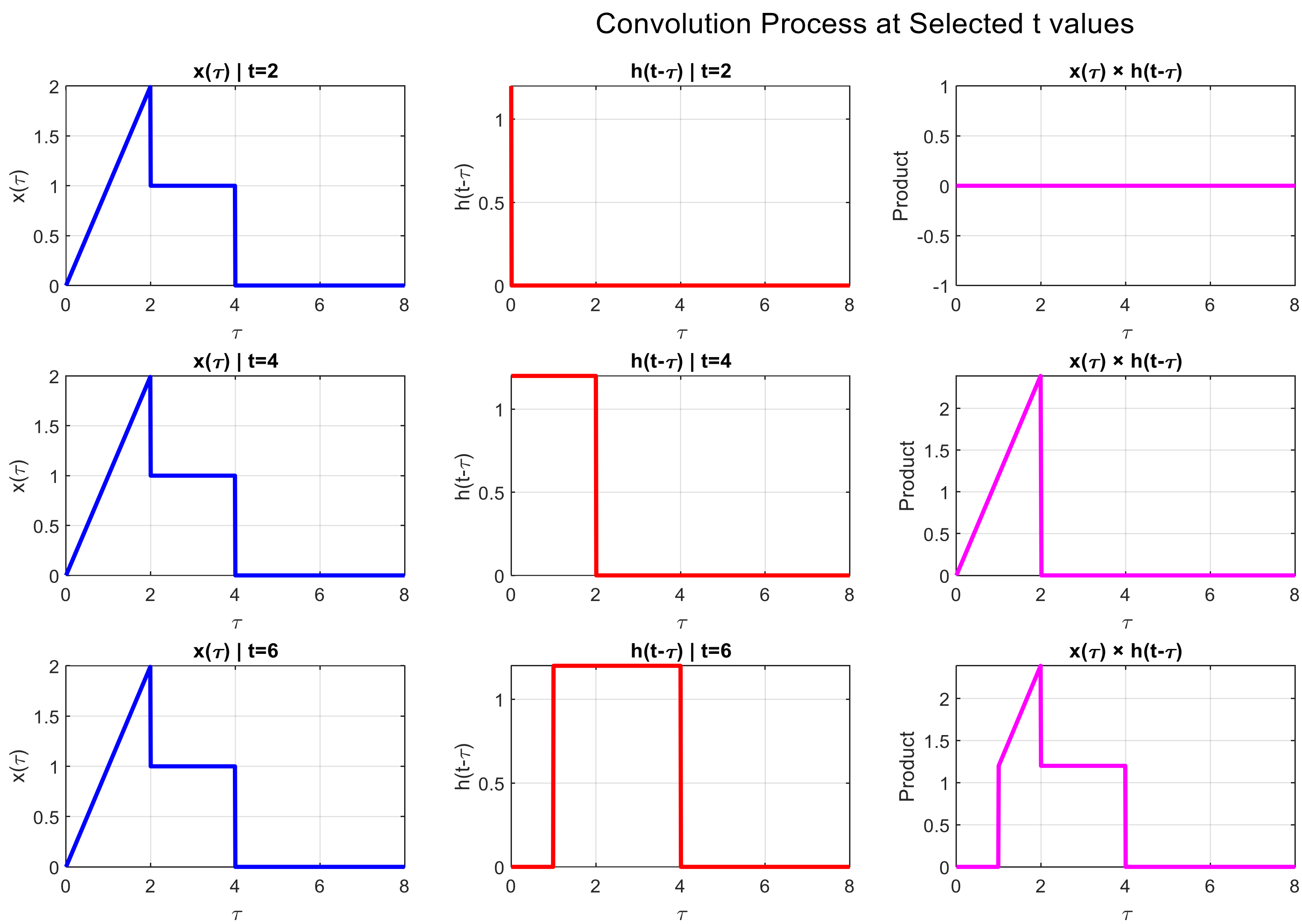
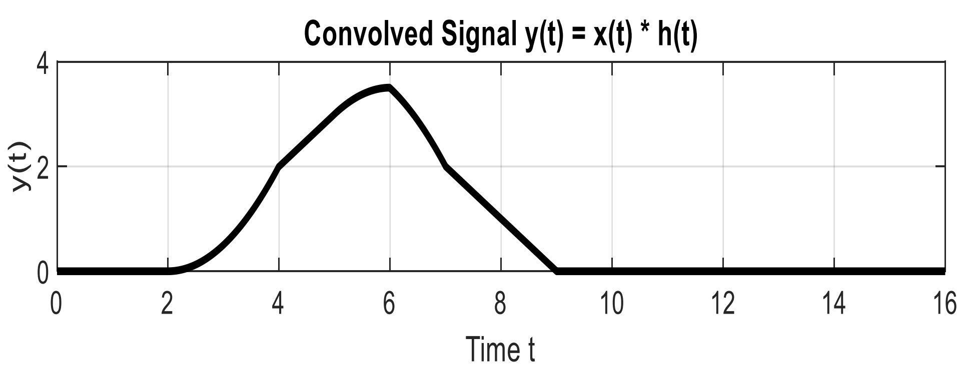


Fig-2

SOLUTION





CLO # 3: Analyze signals using Fourier series representation.

Q3: Analyze the periodic signal shown in Fig-3, calculate all the coefficients of trigonometric Fourier series and also write the final expression of trigonometric Fourier series. **[20 marks]**

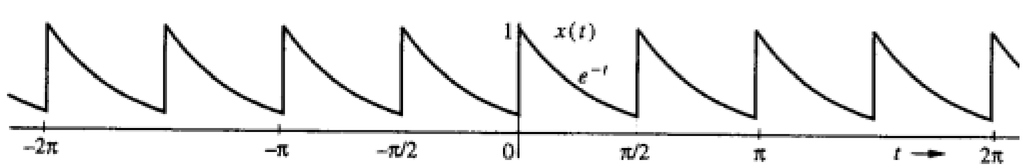


Fig-3

The following mathematical expressions can be used:

$$\int x^2 e^{ax}\,dx = \frac{e^{ax}}{a^3}(a^2x^2 - 2ax + 2)$$

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$$

SOLUTION:

$$T_0 = \pi/2, \quad \omega_0 = \frac{2\pi}{T_0} = 4,$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos 4nt + b_n \sin 4nt$$

$$a_0 = \frac{2}{\pi}\int_0^{\pi/2} e^{-t}\,dt = 0.504.$$

$$a_n = \frac{4}{\pi}\int_0^{\pi/2} e^{-t}\cos 4nt\,dt = \left(\frac{2}{1+16n^2}\right)0.504$$

$$b_n = \frac{4}{\pi}\int_0^{\pi/2} e^{-t}\sin 4nt\,dt = 0.504\left(\frac{8n}{1+16n^2}\right)$$

$$x(t) = 0.504\left[1 + \sum_{n=1}^{\infty}\frac{2}{1+16n^2}\left(\cos 4nt + 4n\sin 4nt\right)\right]$$

CLO # 4: Analyze signals using Fourier transform and its properties.

Q4: Analyze the signal given below, find and sketch the Fourier transform of the modulated signal $x(t)\cos(20t)$ in which $x(t)$ is a gate pulse $rect\left(\frac{t}{8}\right)$ as illustrated in figure given below. **[15 marks]**

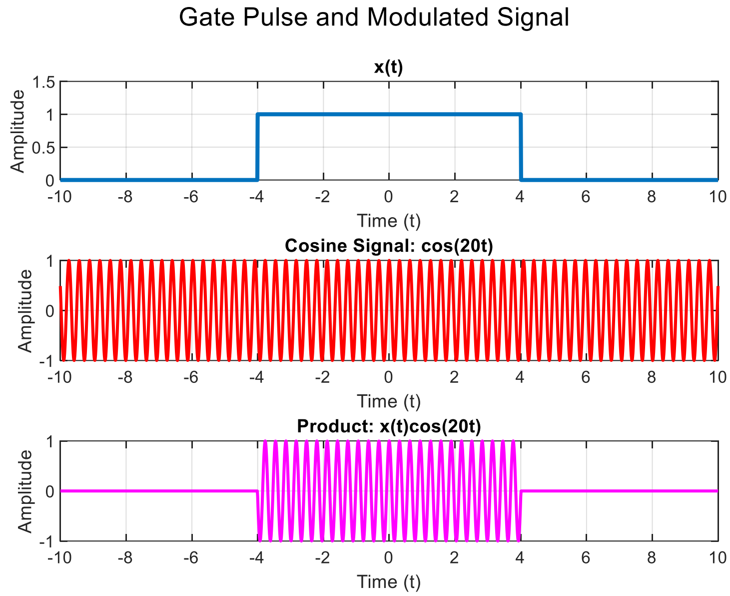


Fig-4

<u>SOLUTION:</u>

Using Frequency shifting property

$$x(t)\cos\omega_0 t = \frac{1}{2}\left[x(t)e^{j\omega_0 t} + x(t)e^{-j\omega_0 t}\right]$$

Hence,

$$x(t)\cos\omega_0 t = \frac{1}{2}\left[X(\omega-\omega_0)+X(\omega+\omega_0)\right] \rightarrow 1$$

And, the spectrum of rectangular Gate Pulse is

$$= \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right).$$

Time period

$$X(\omega) = 8\operatorname{sinc}\left(\frac{\omega 8}{2}\right) = 8\operatorname{sinc}(4\omega)$$

Hence,

$$\operatorname{rect}\left(\frac{t}{8}\right) = 8\operatorname{Sinc}(4\omega) \longrightarrow 2$$

Therefore, Eq 1 becomes

$$x(t)\cos 20t = \frac{1}{2}\left[X(\omega+20)+X(\omega-20)\right]$$

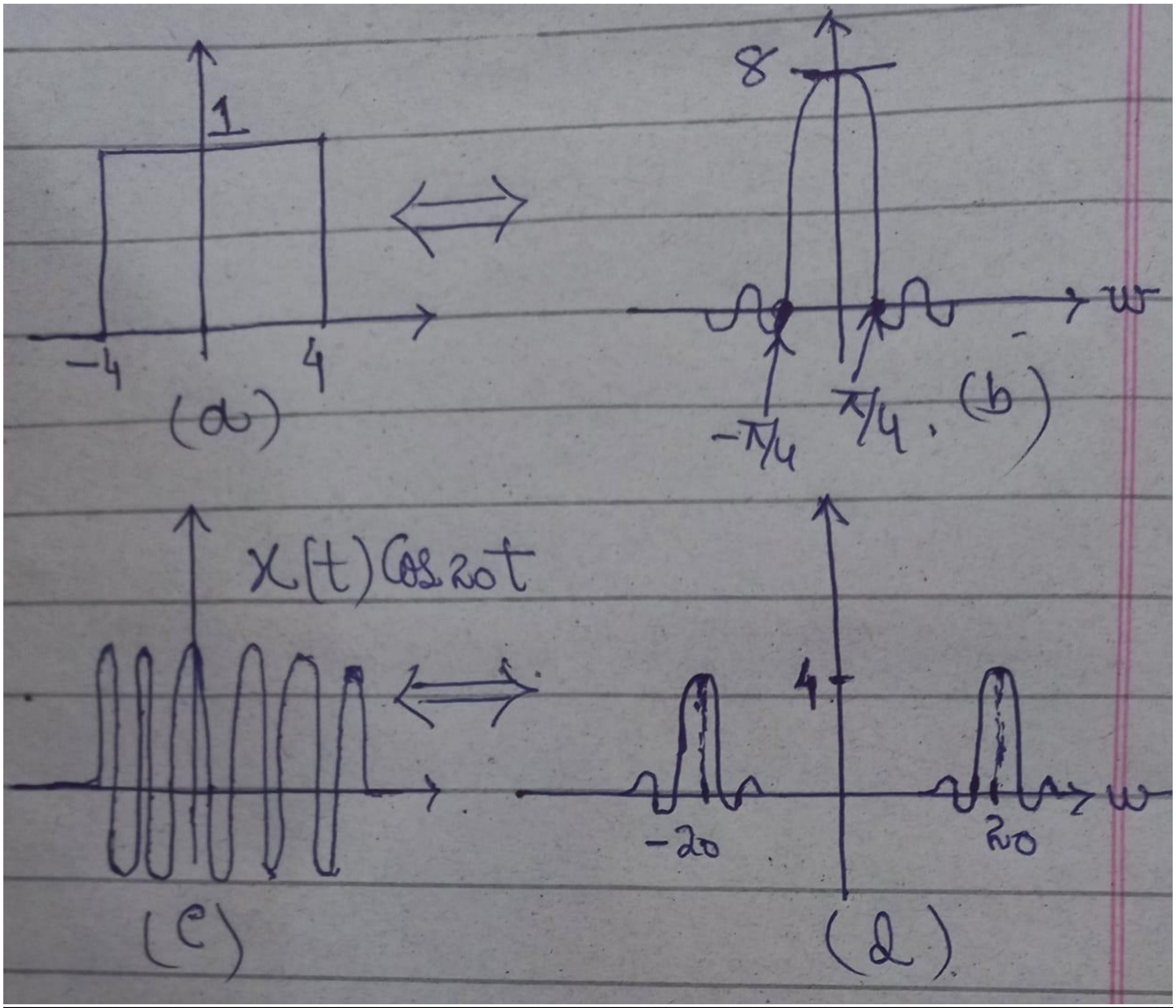
In this Case, $X(\omega) = 8\operatorname{Sinc}(4\omega)$

$$x(t)\cos 20t \Leftrightarrow \frac{1}{2}\left[8\operatorname{Sinc}4(\omega+20)+8\operatorname{Sinc}4(\omega-20)\right]$$

$$x(t)\cos 20t \Leftrightarrow 4\operatorname{Sinc}4(\omega+20)+4\operatorname{Sinc}4(\omega-20)]$$

$$\longrightarrow 3$$

Eq 2 and Eq 3 will used to make Fourier Spectrums.



CLO # 4: Analyze signals using Fourier transform and its properties.

Q5: Analyze and find the Fourier transform of $e^{-|t|}\cos(2t)$ using appropriate property of Fourier transform. **[15 marks]**

SOLUTION:

Find the Fourier transform of

$$x_1(t) = e^{-|t|} \cos(2t)$$

We know $\quad e^{-at} u(t) \Longleftrightarrow \dfrac{1}{a+j\omega}$

so $\quad e^{-t} u(t) \Longleftrightarrow \dfrac{1}{1+j\omega}$

and $\quad e^{-a|t|} \Longleftrightarrow \dfrac{2a}{a^2+\omega^2}$

so $\quad e^{-|t|} \Longleftrightarrow \dfrac{2}{1+\omega^2}$

Also $\quad \cos \omega_0 t \Longleftrightarrow \pi\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]$

so $\quad \cos 2t \Longleftrightarrow \pi\,\delta(\omega-2) + \pi\,\delta(\omega+2)$

Now apply the frequency shifting property

$$x(t)\cos\omega_0 t \Longleftrightarrow \frac{1}{2}\left[X(\omega-\omega_0) + X(\omega+\omega_0)\right]$$

$\cancel{x}$

$$e^{-|t|}\cos(2t) \Longleftrightarrow \frac{1}{1+(\omega-2)^2} + \frac{1}{1+(\omega+2)^2}$$

CLO # 5: Describe and use the sampling theorem.

Q6: A television signal (video and audio) has a bandwidth of 5 MHz. This signal is sampled, quantized, and binary coded to obtain a PCM signal. **[10 marks]**

a) **Describe** what the sampling rate will be if the signal is to be sampled at a rate 25% above the Nyquist rate.

b) If the Nyquist samples are quantized into 1024 levels and then binary coded, **describe** the appropriate number of binary digits required to encode a sample.

c) **Describe** the appropriate number of binary pulse rate (bit/s) required to encode the video and audio signal.

<u>SOLUTION:</u>

<u>(a):</u>

Nyquist Rate $= 2 \times$ Bandwidth

$= 2 \times 5\ (MHz)$.

$= 10 MHz$

Above 25% Nyquist Rate $= \frac{25}{\cancel{100}_{10}} \times \cancel{10}^{1}$

$= 2.5 MHz$.

Required Sampling rate $= 10 + 2.5$

$= 12.5 MHz$

<u>(b):</u>

Bits per Sample $= \log_2 L$

$= \log_2 (1024)$

$= 10$ bits/Sample

<u>(c):</u>

Bit Rate $= 10 \times (12.5 MHz)$

Bit Rate $= 125 MHz$.